

TOP PRIORITY: MIDTERM MASTERY

IIUC CSE • SEMESTER: SPRING 2024 / AUTUMN 2025 / SPRING 2025

30 MARKS • 1:30 HOURS

MATH-1107 / MATH-1101: Mathematics-I

Differential & Integral Calculus — The Ultimate Midterm Examination Preparation Guide & Model Khata Solutions. Engineered from complete course materials, official question sheets, classroom lectures, handwritten notes, and previous year exam scripts.

COURSE CODE & TITLE

MATH-1107 / MATH-1101 •
Mathematics-I

ACADEMIC PROGRAM

B.Sc. in CSE • IIUC

COURSE INSTRUCTOR

Md. Monir Hossain (Lecturer in
Math, CU M.S.)

TARGET EXAMINATION

Mid Term Examination (30 Marks /
90 Mins)



How to Use This Guide

Strategic preparation paths tailored to your remaining preparation time.

Study Framework & Priority Coding

Every single question, concept, and formula in this guide is derived from the official course Question Sheet, 80-page handwritten lecture notebook, CamScanner notes, and past examination papers (Spring 2024, Autumn 2025, Spring 2025). Items are tagged with priority badges:

● Priority A — MUST STUDY

Guaranteed to appear (6M, 4M, 3M problems). 22–26 marks out of 30 come from these exact patterns.

● Priority B — HIGH IMPORTANCE

Alternative questions or high-probability 2M/3M topics (Standard limits, piecewise continuity).

● Priority C — MEDIUM IMPORTANCE

Foundational theorems & definitions (Even/Odd, graph sketches, general definitions).

● Priority D — LOW IMPORTANCE

Supplementary theoretical proofs rarely asked directly.

Time-Budgeted Action Plans:

- 🔴 **If you have 24 Hours (1 Day):** Follow Sections 4 & 5 to understand the frequency pattern, master all formulas in Section 7, thoroughly solve all ● Priority A and ● Priority B model answers, review all ⚠️ Common Mistakes, and write out Mock Exam 1 & 2 under 90-minute timed conditions.
- 🕒 **If you have 3 Hours:** Jump directly to Section 19 (3-Hour Revision Plan). Master: (1) Leibniz Theorem ($y = x^n \ln x$ & $y = a n^{-1} x$), (2) Piecewise Differentiability at boundary points, (3) Circle Related Rates, (4) Rolle's Theorem & Lagrange MVT verification, and (5) First Principle of $\sin x$ & e^x . This secures 24+ marks.
- ⚡ **If you have 1 Hour (Emergency):** Jump directly to Section 20 (1-Hour Emergency Revision). Review the 12 Indeterminate Form answers, the circle problem formula, the two standard Rolle/MVT verification steps, and the Leibniz theorem formulas.



Exam Syllabus & Question Blueprint

Official IIUC CSE Midterm marks distribution, CLO mappings, and question paper anatomy.

Course Learning Outcomes (CLOs)

CLO 1: Compute functions, limits, continuity, differentiability, derivatives, and extrema of single-variable functions.

CLO 2: Understand and apply techniques of successive differentiation, Leibniz's theorem, Rolle's & Mean Value Theorems, series expansions, and related rates.

Question Paper Structure (30 Full Marks • 1:30 Hours)

Question No.	Sub-question	Topic & Focus Area	CLO	Bloom's Level	Marks
Question 1 (Total: 10 Marks)	1(a)	Maclaurin Series Expansion ($f(x) = \log(1+x)$ or $\cos x$)	CLO 1	Understand (U)	2 MARKS
	1(b)	Domain & Range ($ x+3 + x-3 $, $\sqrt{x^2-9}$, or Math input error	CLO 1	Understand (U)	2 MARKS
	1(c)	Differentiability Test at a point (Piecewise functions, with Internal Choice)	CLO 1	Apply (App)	3 MARKS
	1(d)	Indeterminate Form (L'Hôpital) OR Derivative w.r.t Another Function	CLO 1	Understand (U)	3 MARKS
Question 2 (Total: 10 Marks)	2(a)	Differential Co-efficient by First Principle ($\sin x$, e^x , $\ln x$, $\log_a x$)	CLO 2	Understand (U)	3 MARKS
	2(b)	First Derivative: Chain Rule / Logarithmic ($\log \sec(ax+b)^3$, x^{x^x} , nested roots)	CLO 2	Understand (U)	3 MARKS
	2(c)	Related Rates: Circle radius & area rate of change problem	CLO 2	Apply (App)	4 MARKS
Question 3 (Total: 10 Marks)	3(a)	Successive Differentiation / Leibniz's Theorem ($y = x^n \log x$, $y = an^{-1}x$, $e^{a \sin^{-1} x}$)	CLO 2	Apply (App)	6 MARKS
	3(b)	Rolle's Theorem Verification OR Mean Value Theorem Verification	CLO 2	Remember & Understand (R&U)	4 MARKS

Topic Priority Ranking Matrix

Weighted ranking combining PYQ appearance, class emphasis, marks yield, and exam recency.

Priority Level	Topic & Core Problem Pattern	Exam Weightage	Observed Semesters	Preparation Recommendation
 PRIORITY A	Successive Differentiation & Leibniz Theorem • Math input error • $y = an^{-1}x \implies (1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$ • Math input error	6 Marks (Q3a Mandatory)	Sp-24, Au-25, Sp-25	Mandatory 100% Master Highest single-question yield. Practice both 1st derivative squaring and differentiation steps.
 PRIORITY A	Related Rates (Circle Radius & Area) Math input error find Math input error when $r = 5extcm$ and $A = 4\pi extcm^2$.	4 Marks (Q2c Mandatory)	Sp-24, Teacher Notes	Guaranteed 4 Marks. Extremely high probability of repeated identical numbers. Review teacher notes and Sp-24.
 PRIORITY A	Rolle's & Lagrange Mean Value Theorem Verification • Rolle: $f(x) = x^4 - 2x^2$ on $[-2, 2]$; $2x^3 + x^2 - 4x - 2$; $x^2 - 3x + 2$ • MVT: $f(x) = 2x^3 - 8x + 1$ on $[1, 3]$; $3 + 2x - x^2$; Math input error	4 Marks (Q3b Mandatory Choice)	Sp-24, Au-25, Sp-25	Guaranteed 4 Marks. As appears as Q3(b). Memorize the 3 conditions for Rolle's Theorem and 2 conditions for MVT.
 PRIORITY A	Differentiability at a Point (Piecewise Functions) • $f(x) = 3 + 2x/3 - 2x/-3 - 2x$ at $x = 3/2$ or $x = 0$ • $f(x) = 1 + \sin x/2 + (x - \pi/2)^2$ at $x = \pi/2$ • $f(x) = 3x + 2/12 - x^2$ at $x = 2$	3 Marks (Q1c Mandatory Choice)	Sp-24, Sp-25	Must always test Continuity first, then compute Left Hand Derivative (LHD) and Right Hand Derivative (RHD).

Priority Level	Topic & Core Problem Pattern	Exam Weightage	Observed Semesters	Preparation Recommendation
 PRIORITY A	Differential Co-efficient by First Principle • $\sin x$ (Sp-24), e^x (Sp-25), $\ln x$ (Au-25), $\log_a x$, x^n, $e^{\sqrt{x}}$	3 Marks (Q2a Mandatory)	Sp-24, Au-25, Sp-25	Always Q2(a). The limit definition Math input error be explicitly written.
 PRIORITY A	Indeterminate Form Limits (L'Hôpital / Trig / Log) • Math input error • $\lim_{x \rightarrow 0^+} (1/x)^{\tan x} = 1$ • $\lim_{x \rightarrow \infty} (e^x + 1)^{-2/x} = e^{-2}$ • Math input error	3 Marks (Q1d or Choice)	Sp-24, Au-25, Sp-25	Teacher prepared a special 12-problem sheet. Master all 12 problems completely.
 PRIORITY A	Maclaurin's Series Expansion • $f(x) = \log(1+x)$ (Sp-24, Au-25) • $f(x) = \cos x$ (Sp-25)	2 Marks (Q1a Mandatory)	Sp-24, Au-25, Sp-25	Always opens Q1(a). Show crisp calculation of derivative at $x = 0$. Guaranteed 2 marks.
 PRIORITY A	Domain & Range Determination • Absolute value: $f(x) = x+3 + x-3$ (Sp-24) • Radicals: $\sqrt{x^2-9}$ (Au-25), Math input error (Sp-25), $\sqrt{16-x^2}$	2 Marks (Q1b Mandatory)	Sp-24, Au-25, Sp-25	Express Domain in set-builder and interval notation. Express Range clearly by solving in terms of y .
 PRIORITY B	Derivative with Respect to Another Function • Math input error w.r.t Math input error • Math input error w.r.t Math input error	3 Marks (Q1d Choice)	Sp-24, Au-25, Sp-25	Use trigonometric substitution $x = \text{anheta}$ to collapse functions before differentiating. Result is frequently 1 or constant!
 PRIORITY B	Chain Rule & Infinite Nested Functions • $y = \log \sec(ax+b)^3$ • Math input error • Math input error	3 Marks (Q2b)	Sp-24, Au-25, Sp-25	Standard exam favorite. Replace infinite nested terms with y and differentiate implicitly.
 PRIORITY B	Piecewise Continuity Testing at Points • Math input error ($x \neq 0$), 1 ($x = 0$) at $x = 0$ • Linear piecewise at multiple points ($x = 0, 1, 2$)	3–4 Marks	Au-25, Sp-24	Check: (1) $f(a)$ defined, LHL=RHL, (3) $\lim f(x) = L$. Highlight removable discontinuity!
 PRIORITY C	Taylor's Series Expansion • Expand Math input error in ascending powers of $(x-2)$	2–3 Marks	Au-25	Compute successive derivatives evaluated at $a = 2$.
 PRIORITY C	Even / Odd Functions & Limit Existence • $f(x) = \ln(x + \sqrt{1+x^2})$, $x + 1/x$, $1/(1-x^2)$ • Math input error	2–3 Marks	Au-25, Class Notes	Standard definition tests: $f(-x) = -f(x) \implies$ odd; $f(-x) = f(x) \implies$ even.
 PRIORITY D	Theory Proofs & Elementary Graphs • Statement & Proof of Leibniz Theorem by Induction • Graph of $y = \sqrt{x}$, $1/x$, $x + x-1$	2–4 Marks	Class Notebook	Theoretical proofs are supplementary to calculation problems in IIUC CSE mid-semester exams.



Previous Year Question (PYQ) Frequency Database

Exhaustive cross-referencing across Spring 2024, Autumn 2025, and Spring 2025 examinations.

Question Formulation	Topic Section	Semesters Appeared	Times	Marks	Difficulty	Priority
If $y = x^n \log x$, prove that Math input error	Successive Diff. (Leibniz)	Sp-24, Lecture Notes	2	6 Marks	Medium	MUST STUDY
If $y = an^{-1}x$, show $(1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$	Successive Diff. (Leibniz)	Sp-25, Lecture Notes	2	6 Marks	Medium-High	MUST STUDY
If Math input error , show $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$	Successive Diff. (Leibniz)	Au-25, Q-Sheet	2	6 Marks	High	MUST STUDY
The radius of a circle increases at 3extcm/sec . Find rate of change of area when: (i) $r = 5extcm$, (ii) $A = 4\pi extcm^2$	Differentiation (Related Rates)	Sp-24, Lecture Notes, Q-Sheet	3	4 Marks	Easy-Medium	MUST STUDY
Determine whether Rolle's Theorem applies to $f(x) = x^4 - 2x^2$ on $[-2, 2]$ and find all values of c	Rolle's Theorem	Sp-24, Q-Sheet	2	4 Marks	Medium	MUST STUDY
Verify Rolle's Theorem for $f(x) = 2x^3 + x^2 - 4x - 2$ over $[-\sqrt{2}, \sqrt{2}]$	Rolle's Theorem	Sp-25, Q-Sheet	2	4 Marks	Medium	MUST STUDY
Verify Rolle's Theorem for $f(x) = x^2 - 3x + 2$ in $(1, 2)$	Rolle's Theorem	Au-25, Q-Sheet	2	4 Marks	Easy	MUST STUDY
Verify Mean Value Theorem for $f(x) = 2x^3 - 8x + 1$ when $a = 1, b = 3$	Mean Value Theorem	Sp-24, Q-Sheet	2	4 Marks	Medium	MUST STUDY
Verify Mean Value Theorem for $f(x) = 3 + 2x - x^2$ in $(0, 1)$	Mean Value Theorem	Sp-25, Q-Sheet	2	4 Marks	Easy-Medium	MUST STUDY
Verify Mean Value Theorem for Math input error over $[0, 2]$	Mean Value Theorem	Au-25, Q-Sheet	2	4 Marks	Easy-Medium	MUST STUDY
Differentiate $f(x) = \sin x$ by First Principle method	First Principle	Sp-24, Handwritten Notes	2	3 Marks	Easy-Medium	MUST STUDY
Differentiate $f(x) = e^x$ by First Principle method	First Principle	Sp-25, Handwritten Notes	2	3 Marks	Easy	MUST STUDY
Differentiate $f(x) = \ln x$ by First Principle method	First Principle	Au-25, Handwritten Notes	2	3 Marks	Medium	MUST STUDY
Evaluate limit using L'Hôpital: Math input error	Indeterminate Form	Sp-24, Solved Sheet	2	3 Marks	Easy-Medium	MUST STUDY
Evaluate limit using L'Hôpital: $\lim_{x \rightarrow 0^+} (1/x)^{ax}$	Indeterminate Form	Sp-25, Solved Sheet	2	3 Marks	Medium	MUST STUDY
Evaluate limit using L'Hôpital: $\lim_{x \rightarrow \infty} (e^x + 1)^{-2/x}$	Indeterminate Form	Au-25, Solved Sheet	2	3 Marks	Medium	MUST STUDY
Test differentiability of $f(x)$ at $x = 3/2$ ($3 + 2x/3 - 2x/-3 - 2x$)	Differentiability	Sp-24, Lecture Notes	2	3 Marks	Medium	MUST STUDY
Test differentiability of $f(x)$ at $x = \pi/2$ ($1/1 + \sin x/2 + (x - \pi/2)^2$)	Differentiability	Sp-24, Solved Notes	2	3 Marks	Medium	MUST STUDY
Test differentiability of $f(x)$ at $x = 2$ ($3x + 2; x > 2$ and $12 - x^2; x \leq 2$)	Differentiability	Sp-25, Solved Notes	2	3 Marks	Easy-Medium	MUST STUDY

Question Formulation	Topic Section	Semesters Appeared	Times	Marks	Difficulty	Priority
Test differentiability of $f(x)$ at $x = 2$)	Differentiability	Sp-25, Solved Notes	2	3 Marks	Medium	MUST STUDY
Differentiate w.r.t	Diff. w.r.t Function	Sp-24, Sp-25	2	3 Marks	Easy (Substitution)	HIGH IMPORTANC
Differentiate w.r.t	Diff. w.r.t Function	Au-25, Q-Sheet	2	3 Marks	Medium	HIGH IMPORTANC
Find Maclaurin Series for $f(x) = \log(1 + x)$	Maclaurin Series	Sp-24, Au-25	2	2 Marks	Easy	MUST STUDY
Find Maclaurin Series for $f(x) = \cos x$	Maclaurin Series	Sp-25, Q-Sheet	2	2 Marks	Easy	MUST STUDY
Find domain and range of $f(x) = x + 3 + x - 3 $	Domain & Range	Sp-24, Q-Sheet	2	2 Marks	Easy-Medium	MUST STUDY
Find domain and range of $f(x) = \sqrt{x^2 - 9}$	Domain & Range	Au-25, Q-Sheet	2	2 Marks	Easy	MUST STUDY
Find domain and range of	Domain & Range	Sp-25, Q-Sheet	2	2 Marks	Medium	MUST STUDY
Test continuity of $(x \neq 0)$, $1 (x = 0)$ at $x = 0$	Continuity	Au-25, Solved Notes	2	3 Marks	Easy-Medium	HIGH IMPORTANC
If $y = \sqrt{\sin x} + \sqrt{\sin x} + \dots$, show	Nested Differentiation	Sp-25, Q-Sheet	2	3 Marks	Easy	HIGH IMPORTANC
If $y = (\cos x)^{(\cos x)^{\dots}}$, find	Nested Differentiation	Au-25, Q-Sheet	2	3 Marks	Medium	HIGH IMPORTANC
Expand in ascending powers of $(x - 2)$ by Taylor's Series	Taylor's Series	Au-25, Q-Sheet	2	3 Marks	Medium	MEDIUM

🔥 Most Repeated Questions:

- Leibniz Theorem: (Sp-24 + notes)
- Related Rates: Circle area / radius rate of change ($\frac{dr}{dt} = 3, r = 5, A = 4\pi$)
- Maclaurin Series: $\log(1 + x)$ appeared in both Sp-24 and Au-25!
- Inverse trig derivative: w.r.t (Sp-24 & Sp-25)

📚 Most Repeated Mathematical Methods:

- Leibniz Formula Expansion:
- Trigonometric Substitution: $x = \text{anheta}$ to collapse composite inverse angles
- First Principle Limit Definition:
- One-Sided Differentiability Test: vs



Mathematics Topic Hierarchy

Four-level structured dependency roadmap for high-efficiency revision.

LEVEL 1 — MUST MASTER (Guaranteed 24+ Marks)

- Successive Differentiation & Leibniz Theorem: $y = x^n \log x, y = a n^{-1} x, y = e^{a \sin^{-1} x}$.
- Related Rates Applications: Circle area & radius rates.
- Rolle's Theorem & Lagrange MVT: Verifying algebraic polynomials $(x^4 - 2x^2, 2x^3 + x^2 - 4x - 2, 2x^3 - 8x + 1)$ and finding $c \in (a, b)$.
- First Principle Differentiation: $\sin x, e^x, \ln x, \log_a x, x^n, e^{\sqrt{x}}$.

5. **Differentiability of Piecewise Functions:** Boundary value checking ($x = 3/2, 0, 2, \pi/2$).
6. **Indeterminate Forms & L'Hôpital:** Powers taking $\ln$, standard trig limits, **Math input error** and **Math input error**.

LEVEL 2 — HIGH PRIORITY (Alternative & Sectional Marks)

1. **Maclaurin's Series:** $\log(1+x)$, $\cos x$, e^x , $\sin x$.
2. **Domain & Range:** Modulus sums $|x+a| + |x-b|$, square root rationals $\sqrt{(1-x)/x}$, quadratics $\sqrt{x^2-9}$.
3. **Derivative w.r.t Another Function:** Inverse trig forms using $x = \text{antheta}$.
4. **Infinite Nested Derivatives:** $\sqrt{\sin x + \dots}$, $(\cos x)^{(\cos x)^{\dots}}$.
5. **Piecewise Continuity:** Removable and jump discontinuity proofs.

LEVEL 3 — SUPPORTING CONCEPTS

1. **Taylor's Series:** Expanding $1/x$ in powers of $(x-2)$.
2. **Function Classification:** Explicit vs Implicit, One-to-One, Onto, Inverse functions.
3. **Even/Odd Symmetry:** $\ln(x + \sqrt{1+x^2})$, $x + 1/x$.
4. **Limit Existence:** Testing one-sided limits of quotient modulus expressions **Math input error**.

LEVEL 4 — LOW PRIORITY / SUPPLEMENTARY

1. **Graph Sketching:** $y = \sqrt{x}$, $y = 1/x$, $y = -x^2 + 1$, piecewise graphs.
2. **Trigonometric Series Identity:** **Math input error**.
3. **Leibniz Theorem Full Inductive Proof.**



Question Type Classification & Exam Strategy

Algorithmic recognition tricks and standard recipes for each problem type.

Type 1: First Principle Derivations (3M)

Recognize: "By definition / First Principle method, find differential co-efficient..."

Recipe: Write **Math input error**. Use trig compound angle or exponent factoring. Apply standard limit **Math input error** or **Math input error**.

Trap: Skipping the limit symbol before substituting $h = 0$.

Type 2: Domain & Range Calculation (2M)

Recognize: "Find the domain & range of $f(x) = \dots$ "

Recipe: For $\sqrt{E}$, set $E \geq 0$. For P/Q , set $Q \neq 0$. For Range, let $y = f(x)$, express x in terms of y , and determine valid real y values.

Trap: Writing range as $\mathbb{R}$ for radicals instead of $[0, \infty)$.

Type 3: Piecewise Continuity / Differentiability (3M)

Recognize: Multi-branch piecewise function tested at a partition point $x = c$.

Recipe: Step 1: Evaluate $f(c)$. Step 2: Compute LHL & RHL. If equal, continuous. Step 3: Compute LHD & RHD. If continuous AND LHD=RHD, differentiable.

Trap: Checking LHD=RHD without testing continuity! (If discontinuous, immediately not differentiable).

Type 4: Indeterminate Form Limits (3M)

Recognize: Limit resulting in $0/0, \infty/\infty, 0 \cdot \infty, 1^\infty, \infty^0$.

Recipe: For $f(x)^{g(x)}$, let $y = f(x)^{g(x)}$, take $\ln y = g(x) \ln f(x)$, rewrite as quotient, apply L'Hôpital, then $L = e^{\lim \ln y}$. For trig, simplify before differentiating.

Trap: Forgetting to exponentiate back ($L = e^b$) after finding the limit of $\ln y$.

Type 5: Related Rates Word Problem (4M)

Recognize: "Radius increases at rate of... find rate of change of area..."

Recipe: Write $A = \pi r^2$. Differentiate w.r.t t . **Math input error**. Substitute given **Math input error** and r . If Area given, solve for $r = \sqrt{A/\pi}$ first.

Trap: Forgetting units ($\text{extcm}^2/\text{extsec}$) in the final answer.

Type 6: Leibniz Theorem Proofs (6M)

Recognize: Differential equations involving y_{n+2}, y_{n+1}, y_n .

Recipe: Differentiate once: get y_1 . Clear denominator/square both sides. Differentiate again: divide out $2y_1$ to get relation between y_2, y_1, y . Apply Leibniz's theorem n times.

Trap: Errors in binomial coefficients **Math input error** and **Math input error**.

Type 7: Rolle's & MVT Verification (4M)

Type 8: Maclaurin & Taylor Expansions (2–3M)

Recognize: "Verify Rolle's / Mean Value Theorem for $f(x)$ on $[a, b]$..."

Recipe: State continuity on $[a, b]$ (polynomial). State differentiability on (a, b) . For Rolle, check $f(a) = f(b)$, then solve $f'(c) = 0$. For MVT, solve $f'(c) = \frac{f(b) - f(a)}{b - a}$. Confirm $c \in (a, b)$.

Trap: Selecting a root for c that lies outside the open interval (a, b) .

Recognize: "Obtain Maclaurin series generated by $f(x) = \dots$ "

Recipe: Compute $f(0), f'(0), f''(0), f'''(0), f^{(4)}(0)$. Substitute into $\sum \frac{f^{(n)}(0)}{n!} x^n$.

Trap: Sign errors when taking repeated derivatives of $\log(1+x)$.



Formula Master Sheet (Math Formula Bank)

Comprehensive reference bank with recognition tricks, common mistakes, and Bangla memory anchors.

Category 1: First Principle (Definition of Derivative)

1.1 Fundamental First Principle Definition

MUST MEMORIZE

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Meaning:

Computes the instantaneous rate of change of $f(x)$ at any point x as the step h approaches zero.

Recognition Trick:

Look for the phrase "by first principle" or "by definition". Never use direct power/chain rule here!

Variables:

$f(x)$ = original function, h = infinitesimal increment in x , $f'(x)$ = first derivative.

Common Mistake:

Dropping the limit symbol $\lim_{h \rightarrow 0}$ in intermediate steps or setting $h = 0$ too early causing division by zero.

When to Use:

Whenever the question explicitly demands: "By definition" or "From first principle, differentiate..."

বাংলায় মনে রাখার উপায়:

ফাংশনে x -এর জায়গায় $x + h$ বসাবো, মূল ফাংশন বিয়োগ করে h দিয়ে ভাগ করবো এবং $h \rightarrow 0$ লিমিট বের করবো।

1.2 First Principle of Power Function

PRIORITY A

$$\frac{d}{dx}(x^n) = n x^{n-1} \quad \text{via} \quad \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} = x^n \lim_{h \rightarrow 0} \left[\frac{n}{x} + \frac{n(n-1)}{2!} \frac{h}{x^2} + \dots \right]$$

When to Use:

When asked to derive $\frac{d}{dx}(x^n)$ by definition.

Recognition Trick:

Factor out x^n from $(x+h)^n$ as $x^n(1+h/x)^n$ and apply Binomial Theorem.

Common Mistake:

Forgetting to factor h from the numerator before canceling with the denominator h .

বাংলায়:

$(x+h)^n$ থেকে x^n কমন নিয়ে দ্বিপদী বিস্তৃতি (Binomial expansion) করে h কেটে দিয়ে $h = 0$ বসাতে হয়।

1.3 First Principle of Exponential Functions

PYQ (SP-25)

$$\frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(e^{mx}) = m e^{mx}, \quad \frac{d}{dx}(e^{\sqrt{x}}) = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$$

Key Standard Limit:

$\lim_{u \rightarrow 0} \frac{e^u - 1}{u} = 1$. Factor e^x out of the numerator: $e^{x+h} - e^x = e^x(e^h - 1)$.

When to Use:

Deriving exponential derivatives by definition.

Common Mistake:

For $e^{\sqrt{x}}$, failing to rationalize the exponent difference $\sqrt{x+h} - \sqrt{x}$ by multiplying numerator and denominator by $\sqrt{x+h} + \sqrt{x}$.

বাংলায়:

e^x কমন নিলে ব্র্যাকেটে $(e^h - 1)/h$ থাকে, যার মান সরাসরি 1। আর $e^{\sqrt{x}}$ -এর বেলায় অনুবন্ধী (conjugate) দিয়ে গুণ করতে হয়।

1.4 First Principle of Logarithmic Functions

PYQ (AU-25)

$$\frac{d}{dx}(\ln x) = \frac{1}{x}, \quad \frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

Key Log Property:

$\log_a(x+h) - \log_a x = \log_a\left(\frac{x+h}{x}\right) = \log_a\left(1 + \frac{h}{x}\right)$. Deriving $\frac{d}{dx} \ln x$ (Au-25) or $\log_a x$ by definition.

When to Use:

Deriving $\frac{d}{dx} \ln x$ (Au-25) or $\log_a x$ by definition.

Common Mistake:

Forgetting the change-of-base factor $\frac{1}{\ln a}$ when base is a instead of e .

বাংলায়:

লগের বিয়োগ মানে ভাগের সূত্র:

$\ln(x+h) - \ln x = \ln(1+h/x)$ । নিচে x দিয়ে গুণ-ভাগ করলেই $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$ হয়ে যায়।

1.5 First Principle of Trigonometric Functions

PYQ (SP-24)

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x, \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

Key Trig Identity:

$\sin(x+h) - \sin x = 2 \cos\left(x + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right)$, or $\sin(x+h) = \sin x \cos h + \cos x \sin h$.

When to Use:

Finding derivative of $\sin x$ (asked in Sp-24 exam for 3 marks!).

Common Mistake:

Writing $\lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 1$ instead of 0! Note: $\lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0$, whereas $\frac{1 - \cos h}{h^2} = \frac{1}{2}$.

বাংলায়:

$\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$ সূত্রে

ফেললে $\sin(h/2)/(h/2) \rightarrow 1$ হয় এবং

$\cos(x+h/2) \rightarrow \cos x$ থাকে।

Category 2: Related Rates (Circle Area & Radius)

$$A = \pi r^2 \implies \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

Meaning:

The time rate of change of a circle's area is the circumference ($2\pi r$) multiplied by the radial expansion rate ($\frac{dr}{dt}$).

Variables:

A = area (cm^2), r = radius (cm), t = time (sec),
 $\frac{dr}{dt} = 3 \text{ cm/s}$ (given).

Case (i) $r = 5 \text{ cm}$ Solution:

$$\frac{dA}{dt} = 2\pi(5)(3) = 30\pi \text{ cm}^2/\text{sec} \approx 94.25 \text{ cm}^2/\text{sec}$$

Case (ii) $A = 4\pi \text{ cm}^2$ Solution:

$$\pi r^2 = 4\pi \implies r^2 = 4 \implies r = 2 \text{ cm}$$

$$\frac{dA}{dt} = 2\pi(2)(3) = 12\pi \text{ cm}^2/\text{sec} \approx 37.70 \text{ cm}^2/\text{sec}$$

Common Mistake:

Plugging $A = 4\pi$ directly in place of r without solving for radius $r = 2$ first!

বাংলায়:

ক্ষেত্রফল $A = \pi r^2$ -কে সময় t -এর সাপেক্ষে ডিফারেনশিয়েট করলে চেইন রুলে $2\pi r \frac{dr}{dt}$ আসে।
 ক্ষেত্রফল দেওয়া থাকলে আগে ব্যাসার্ধ r বের করে নিতে হবে!

Category 3: Indeterminate Forms & 10 Essential Standard Limits

3.1 L'Hôpital's Rule & Indeterminate Types

CORE RULE

$$\text{If } \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{0}{0} \text{ or } \frac{\infty}{\infty}, \text{ then } \lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

7 Indeterminate Forms:

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \cdot \infty, \infty - \infty, 0^0, 1^\infty, \infty^0$$

Power Form Strategy ($0^0, 1^\infty, \infty^0$):

Let $y = f(x)^{g(x)} \implies \ln y = g(x) \ln f(x) = \frac{\ln f(x)}{1/g(x)}$
 Find limit k , then $L = e^k$.

Common Mistake:

Applying quotient rule $\frac{f'g - fg'}{g^2}$ instead of differentiating numerator and denominator separately!

বাংলায়:

ল' হসপিটালে ওপরেরটা ওপরে ডিফারেনশিয়েট, নিচেরটা নিচে ডিফারেনশিয়েট করতে হয় (ভাগ সূত্র নয়!)। পাওয়ার থাকলে উভয়পাশে $\ln$ ধরে নিতে হবে।

No.	Standard Limit Formula	Indeterminate Type	Value	Bangla Quick Memory Trick
1	$\lim_{x \rightarrow 0} \frac{\sin x}{x}$	$0/0$	1	কোণ আর হর সমান হলে সীমা ১
2	$\lim_{x \rightarrow 0} \frac{\tan x}{x}$	$0/0$	1	$\tan x$ এবং x ক্ষুদ্র হলে অনুপাত ১
3	$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$	$0/0$	$1/2$	$2 \sin^2(x/2)/x^2 \implies 1/2$ (পরীক্ষার ভুল: ১ লিখবেন না!)
4	$\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$	$0/0$	1	এক্সপোনেনশিয়াল বিয়োগ ১ বাই x সমান ১
5	$\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$	$0/0$	$\ln a$	যেকোনো বেস a -এর জন্য $\ln a$
6	$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$	$0/0$	1	লগারিদমের আদর্শ লিমিট
7	$\lim_{x \rightarrow 0} (1+x)^{1/x}$	1^∞	e	অয়লার ধ্রুবক e সংজ্ঞায়িত লিমিট
8	$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$	1^∞	e	অসীম অভিমুখী e -এর মান
9	$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} x^2 \log(x^2)$	$0 \cdot (-\infty)$	0	বহুপদী x লগারিদমের চেয়ে দ্রুত শূন্যে যায়

No.	Standard Limit Formula	Indeterminate Type	Value	Bangla Quick Memory Trick
10	$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$	0/0	1/2	টেলর বিস্তৃতি থেকে দ্বিতীয় পদের সহগ

Category 4: Successive Differentiation & Leibniz's Theorem

4.1 Leibniz's Theorem Formula

 PYQ GUARANTEED 6M

$$(uv)_n = u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2 + \dots + \binom{n}{r} u_{n-r} v_r + \dots + u v_n$$

Meaning:

Expresses the n -th derivative of a product of two functions $u(x)$ and $v(x)$ in terms of their individual derivatives, analogous to the Binomial Theorem.

Binomial Coefficients:

$$\binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}, \quad \binom{n}{3} = \frac{n(n-1)(n-2)}{6}$$

Choice of u and v :

Always choose $v(x)$ as the polynomial factor whose higher derivatives eventually vanish (e.g. $v = 1 - x^2$, so $v_1 = -2x$, $v_2 = -2$, $v_3 = 0$). Choose u as the function you can differentiate n times!

Common Mistake:

Forgetting the minus sign when

$$v = 1 - x^2 \implies v_1 = -2x \text{ and } v_2 = -2.$$

বাংলায়:

যাকে কয়েকবার ডিফারেনশিয়েট করলে শূন্য হয়ে যায় (যেমন $1 - x^2$ বা $x^2 + 1$), তাকে v ধরতে হয়, যাতে n বার ডিফারেনশিয়েট করলে v শূন্য হয়ে যায়।

4.2 Standard n -th Derivative Formulas

 HIGH PRIORITY

$$y = \ln(ax+b) \implies y_n = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}, \quad y = \frac{1}{ax+b} \implies y_n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$y = \sin(ax+b) \implies y_n = a^n \sin\left(ax+b + \frac{n\pi}{2}\right), \quad y = \cos(ax+b) \implies y_n = a^n \cos\left(ax+b + \frac{n\pi}{2}\right)$$

When to Use:

Finding direct n -th derivatives or applying inside Leibniz expansion.

Common Mistake:

Writing $(-1)^n$ for $\ln(ax+b)$ instead of $(-1)^{n-1}$, and writing $n!$ instead of $(n-1)!$.

বাংলায়:

ত্রিকোণমিতিক ফাংশনে প্রতিবার ডিফারেনশিয়েট করলে ভেতরে কোণ $\pi/2$ বাড়ে এবং সামনে a গুণ হয়।

Category 5: Rolle's & Mean Value Theorems

5.1 Rolle's Theorem

PYQ GUARANTEED 4M

$$f(a) = f(b) \implies \exists c \in (a, b) \quad \text{such that} \quad f'(c) = 0$$

Three Hypotheses:

1. $f(x)$ is continuous on $[a, b]$.
2. $f(x)$ is differentiable on (a, b) .
3. $f(a) = f(b)$.

Geometric Interpretation:

There is at least one point c between a and b where the tangent to the curve is horizontal (parallel to the x -axis).

Common Mistake:

Picking a value of c that equals an endpoint ($c = a$ or $c = b$) or lies outside (a, b) . c MUST lie strictly inside (a, b) .

বাংলায়:

দুই প্রান্তবিন্দুতে মান সমান হলে ($f(a) = f(b)$),
মধ্যে অন্তর্গত একটি বিন্দু c থাকবে যেখানে স্পর্শক
সমান্তরাল হবে অর্থাৎ $f'(c) = 0$ ।

5.2 Lagrange's Mean Value Theorem (MVT)

PYQ GUARANTEED 4M

$$\exists c \in (a, b) \quad \text{such that} \quad f'(c) = \frac{f(b) - f(a)}{b - a}$$

Two Hypotheses:

1. $f(x)$ is continuous on $[a, b]$.
2. $f(x)$ is differentiable on (a, b) .

Geometric Interpretation:

There exists at least one point c where the instantaneous slope (tangent) equals the average rate of change (secant line connecting $(a, f(a))$ and $(b, f(b))$).

Common Mistake:

Arithmetic errors when calculating $f(b)$ and $f(a)$ or solving the resulting quadratic equation for c .

বাংলায়:

গড় ঢাল $\frac{f(b)-f(a)}{b-a}$ -এর মান কোনো একটি স্পর্শকের
ঢাল $f'(c)$ -এর সমান হবেই।

Category 6: Series Expansions (Maclaurin & Taylor)

6.1 Maclaurin's Series Formula

PYQ GUARANTEED 2M

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

Standard Expansions:

- $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$
- $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$
- $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

When to Use:

Q1(a) in every IUC midterm exam!

Common Mistake:

Writing factorials in the denominator of $\log(1+x)!$ (It is $x^2/2$, not $x^2/2!$, because the $(n-1)!$ in the numerator cancels $n!$ down to n).

বাংলায়:

$x = 0$ বিন্দুতে বারবার ডিফারেনশিয়েট করে
মানগুলো ম্যাকলরিনের সূত্রে বসিয়ে দেওয়া।

$$f(x) = f(a) + (x - a) f'(a) + \frac{(x - a)^2}{2!} f''(a) + \frac{(x - a)^3}{3!} f'''(a) + \dots$$

When to Use:

Expanding a function in powers of $(x - a)$, e.g.,
expand $\frac{1}{x}$ about $(x - 2)$.

Common Mistake:

Evaluating derivatives at $x = 0$ instead of $x = a$.

বাংলায়:

$(x - a)$ -এর ঘাতে প্রকাশ করতে বললে
ডিফারেনশিয়েট করে $x = a$ বসিয়ে সূত্রে সাজাতে
হয়।



Exam-Khata Model Answers: Part 1

Domain & Range, Limits, Continuity, and Differentiability with exact university khata layout.

Topic 1: Domain & Range Determination

● PYQ (SP-24)

2 MARKS

Find the domain and range of the function $f(x) = |x + 3| + |x - 3|$.

EASY-MEDIUM



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = |x + 3| + |x - 3|$ **Required:** Find the Domain and Range of $f(x)$.

Step 1: Determine the Domain

The absolute value functions $|x + 3|$ and $|x - 3|$ are defined for every real number x . There are no fractions with denominators or square roots of negative numbers.

∴ The function $f(x)$ yields real values for all $x \in \mathbb{R}$.

$$\text{Domain } D_f = \mathbb{R} = (-\infty, \infty)$$

Step 2: Express $f(x)$ Piecewise to Find the Range

The partition points are $x = -3$ and $x = 3$.

1. For $x < -3$: $|x + 3| = -(x + 3)$ and $|x - 3| = -(x - 3)$

$$f(x) = -(x + 3) - (x - 3) = -x - 3 - x + 3 = -2x$$

Since $x < -3$, $-2x > 6$. So $f(x) \in (6, \infty)$.

2. For $-3 \leq x \leq 3$: $|x + 3| = x + 3$ and $|x - 3| = -(x - 3)$

$$f(x) = (x + 3) - (x - 3) = x + 3 - x + 3 = 6$$

3. For $x > 3$: $|x + 3| = x + 3$ and $|x - 3| = x - 3$

$$f(x) = (x + 3) + (x - 3) = 2x$$

Since $x > 3$, $2x > 6$. So $f(x) \in (6, \infty)$.

● সহজ বাংলায় বুঝি

মডুলাস ফাংশনে x -এর যেকোনো বাস্তব মান বসানো যায়, তাই ডোমেন হলো সকল বাস্তব সংখ্যা $\mathbb{R}$ । আর -3 থেকে 3 -এর মধ্যে ফাংশনটির মান স্থির থাকে 6 । এর বাইরে মান কেবল বাড়তেই থাকে। তাই সর্বনিম্ন মান 6 এবং রেঞ্জ হলো $[6, \infty)$ ।

▲ Common Exam Mistake

অনেক শিক্ষার্থী ভুল করে রেঞ্জও $(-\infty, \infty)$ লিখে ফেলে। কিন্তু মডুলাসের যোগফল কখনো ঋণাত্মক হতে পারে না এবং এখানে মান কখনো 6 -এর নিচে নামে না!

● PYQ (AU-25)

2 MARKS

Find the domain and range of the function $f(x) = \sqrt{x^2 - 9}$.

EASY



● PYQ (SP-25)

2 MARKS

Find the domain and range of the function $f(x) = \sqrt{\frac{1-x}{x}}$.

MEDIUM



Topic 2: Limits & Continuity of Functions

Definition of Continuity at a Point (Dual Exam Formats)

A. Bullet-Point Exam Format (Easy to Memorize):

A function $f(x)$ is continuous at $x = a$ if and only if:

- **Condition 1:** $f(a)$ exists (the function is defined at $x = a$).
- **Condition 2:** $\lim_{x \rightarrow a} f(x)$ exists (Left-Hand Limit = Right-Hand Limit, i.e., $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$).
- **Condition 3:** $\lim_{x \rightarrow a} f(x) = f(a)$ (the limit value equals the functional value).

B. Continuous Paragraph Format (Exam Answer-Script Style):

A function $f(x)$ is said to be continuous at a point $x = a$ if the limit of $f(x)$ as x approaches a exists and is equal to the functional value $f(a)$. Formally, this requires that $f(a)$ is well-defined, $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$, and their common value equals $f(a)$. Geometrically, this means the graph of $f(x)$ passes through $(a, f(a))$ without any break, jump, or hole.

PYQ (AU-25)

3 MARKS

Test the continuity of $f(x) = \begin{cases} \frac{1-\cos x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $x = 0$.

EASY-MEDIUM



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = \begin{cases} \frac{1-\cos x}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $x = 0$.Required: Test the continuity of $f(x)$ at $x = 0$.Step 1: Check if $f(0)$ existsFrom the given definition, at $x = 0$:

$$f(0) = 1 \quad (\text{exists and is finite})$$

Step 2: Evaluate $\lim_{x \rightarrow 0} f(x)$ For $x \neq 0$, $f(x) = \frac{1-\cos x}{x^2}$.

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = \left[\frac{1-1}{0} = \frac{0}{0} \text{ form} \right]$$

Method (Trigonometric Identity):

$$\lim_{x \rightarrow 0} \frac{2 \sin^2(x/2)}{x^2} = 2 \lim_{x \rightarrow 0} \left[\frac{\sin(x/2)}{x/2} \right]^2 \cdot \frac{1}{4} = 2 \cdot (1)^2 \cdot \frac{1}{4} = \frac{1}{2}$$

(Alternatively, by L'Hôpital's Rule: $\lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2}$).Therefore, $\lim_{x \rightarrow 0} f(x) = \frac{1}{2}$ exists.

Step 3: Compare Limit with Functional Value

Here, $\lim_{x \rightarrow 0} f(x) = \frac{1}{2}$, whereas $f(0) = 1$.

$$\lim_{x \rightarrow 0} f(x) \neq f(0) \quad \left(\frac{1}{2} \neq 1 \right)$$

● সহজ বাংলায় বুঝি

এখানে $f(0) = 1$ দেওয়া আছে। কিন্তু লিমিট বের করলে দেখা যায় $\lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$ । যেহেতু লিমিটের মান ($1/2$) এবং ফাংশনের মান (1) সমান নয়, তাই ফাংশনটি $x = 0$ বিন্দুতে অবিচ্ছিন্ন (continuous) নয়! এটি একটি বিচ্ছিন্ন ফাংশন (removable discontinuity)।

⚠ Common Exam Mistake (Captured from Handwritten Notes)

In preliminary notes, students frequently calculate $2 \cdot 1 \cdot \frac{1}{4} = \frac{1}{2}$, but accidentally write 1 in the next step and falsely declare it continuous! Always check that $\lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = 1/2$.

PYQ (SP-24)

3 MARKS

Test continuity of $f(x) = \begin{cases} 3+2x, & -3/2 \leq x < 0 \\ 3-2x, & 0 \leq x < 3/2 \\ -3-2x, & x \geq 3/2 \end{cases}$ at $x = 0$ and $x = 3/2$.

MEDIUM



Theorem: Differentiability Implies Continuity

Statement: If a function $f(x)$ is differentiable at a point $x = a$, then it is necessarily continuous at $x = a$. However, the converse is not true (continuity does not guarantee differentiability).

Proof: We can write $f(x) - f(a) = \frac{f(x)-f(a)}{x-a} \cdot (x-a)$ for $x \neq a$.

Taking limit as $x \rightarrow a$:

$$\begin{aligned}\lim_{x \rightarrow a} [f(x) - f(a)] &= \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a} \right] \cdot \lim_{x \rightarrow a} (x - a) = f'(a) \cdot 0 = 0 \\ \implies \lim_{x \rightarrow a} f(x) &= f(a)\end{aligned}$$

Hence, $f(x)$ is continuous at $x = a$.

Counter-example for converse: $f(x) = |x|$ is continuous at $x = 0$, but its LHD is -1 and RHD is $+1$, so it is not differentiable at $x = 0$.

Test the differentiability of $f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \pi/2 \\ 2 + (x - \pi/2)^2, & x \geq \pi/2 \end{cases}$ at $x = \pi/2$.



MODEL EXAM-KHATA SOLUTION

Given: $f(x)$ piecewise function around $x = \pi/2$.

Required: Test differentiability at $x = \pi/2$.

Step 1: Test Continuity at $x = \pi/2$ (Mandatory Prerequisite)

1. Functional value: For $x \geq \pi/2$, $f(x) = 2 + (x - \pi/2)^2$:

$$f(\pi/2) = 2 + (\pi/2 - \pi/2)^2 = 2 + 0 = 2$$

2. Left-Hand Limit: For $0 \leq x < \pi/2$, $f(x) = 1 + \sin x$:

$$\lim_{x \rightarrow (\pi/2)^-} f(x) = 1 + \sin(\pi/2) = 1 + 1 = 2$$

3. Right-Hand Limit: For $x \geq \pi/2$, $f(x) = 2 + (x - \pi/2)^2$:

$$\lim_{x \rightarrow (\pi/2)^+} f(x) = 2 + 0 = 2$$

Since $\text{LHL} = \text{RHL} = f(\pi/2) = 2$, $f(x)$ is continuous at $x = \pi/2$.

Step 2: Test Left-Hand Derivative (LHD) and Right-Hand Derivative (RHD)

Left-Hand Derivative ($Lf'(\pi/2)$):

Using $f(x) = 1 + \sin x$ for $x < \pi/2$:

$$\begin{aligned} Lf'(\pi/2) &= \lim_{h \rightarrow 0^-} \frac{f(\pi/2 + h) - f(\pi/2)}{h} = \lim_{h \rightarrow 0^-} \frac{[1 + \sin(\pi/2 - h)] - 2}{-h} \quad (\text{where } h > 0) \\ &= \lim_{h \rightarrow 0} \frac{1 + \cos h - 2}{-h} = \lim_{h \rightarrow 0} \frac{\cos h - 1}{-h} = \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0 \end{aligned}$$

(Or by derivative of branch: $\frac{d}{dx}(1 + \sin x)|_{x=\pi/2} = \cos(\pi/2) = 0$).

Right-Hand Derivative ($Rf'(\pi/2)$):

Using $f(x) = 2 + (x - \pi/2)^2$ for $x \geq \pi/2$:

$$Rf'(\pi/2) = \lim_{h \rightarrow 0^+} \frac{f(\pi/2 + h) - f(\pi/2)}{h} = \lim_{h \rightarrow 0^+} \frac{[2 + (h)^2] - 2}{h} = \lim_{h \rightarrow 0^+} \frac{h^2}{h} = \lim_{h \rightarrow 0^+} h = 0$$

(Or by derivative of branch: $\frac{d}{dx}[2 + (x - \pi/2)^2]|_{x=\pi/2} = 2(x - \pi/2)|_{x=\pi/2} = 0$).

Step 3: Compare LHD and RHD

$$Lf'(\pi/2) = Rf'(\pi/2) = 0$$

● সহজ বাংলায় বুঝি

প্রথমে দেখলাম $\pi/2$ -তে ফাংশনটি অবিচ্ছিন্ন (মান ২)। এরপর বাম দিকের অন্তরজ $\cos(\pi/2) = 0$ এবং ডান দিকের অন্তরজ $2(x - \pi/2) = 0$ । যেহেতু দুই পাশের অন্তরজ সমান এবং মান ০, তাই ফাংশনটি $\pi/2$ বিন্দুতে অন্তরীকরণযোগ্য (differentiable) এবং $f'(\pi/2) = 0$ ।

Test the differentiability of $f(x) = \begin{cases} 3x + 2, & x > 2 \\ 12 - x^2, & x \leq 2 \end{cases}$ at $x = 2$.



PYQ (SP-25)

3 MARKS

Test the differentiability of $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2-x, & 1 \leq x \leq 2 \\ x - \frac{1}{2}x^2, & x > 2 \end{cases}$ at $x = 2$.

MEDIUM



Exam-Khata Model Answers: Part 2

Indeterminate Forms, First Principles, Advanced Differentiation, and Related Rates.

Topic 4: Indeterminate Forms & L'Hôpital's Rule (All 12 Canonical Teacher Problems)

PYQ (SP-24)

3 MARKS

Problem 10: Evaluate the limit using L'Hôpital's rule: $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

EASY-MEDIUM



MODEL EXAM-KHATA SOLUTION

Given: $L = \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$

Form Check: As $x \rightarrow 0$, $\tan 0 - \sin 0 = 0$ and $0^3 = 0$. Form is $\left[\frac{0}{0}\right]$.

Method 1: By L'Hôpital's Rule (Exam Requirement)

Differentiating numerator and denominator with respect to x :

$$L = \lim_{x \rightarrow 0} \frac{\sec^2 x - \cos x}{3x^2} \quad \left[\frac{1-1}{0} = \frac{0}{0} \text{ form} \right]$$

Differentiating again with respect to x :

$$L = \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x - (-\sin x)}{6x} = \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x + \sin x}{6x}$$

Splitting into standard limits as $x \rightarrow 0$:

$$L = \frac{2}{6} \lim_{x \rightarrow 0} \left(\sec^2 x \cdot \frac{\tan x}{x} \right) + \frac{1}{6} \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Since $\lim_{x \rightarrow 0} \sec x = 1$, $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$, and $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$:

$$L = \frac{1}{3} (1 \cdot 1) + \frac{1}{6} (1) = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2}$$

Method 2: Trigonometric Identity (Quick Verification)

$$\tan x - \sin x = \sin x \left(\frac{1}{\cos x} - 1 \right) = \frac{\sin x (1 - \cos x)}{\cos x}$$

$$L = \lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right] \cdot \left[\frac{1 - \cos x}{x^2} \right] \cdot \left[\frac{1}{\cos x} \right] = 1 \cdot \frac{1}{2} \cdot 1 = \frac{1}{2}$$

● সহজ বাংলায় বুঝি

সরাসরি 0 বসলে 0/0 আসে। তাই ল' হসপিটাল নিয়মে দুইবার ডিফারেনশিয়েট করে স্ট্যান্ডার্ড লিমিট $\frac{\tan x}{x} \rightarrow 1$ এবং $\frac{\sin x}{x} \rightarrow 1$ ব্যবহার করলেই মান চলে আসে 1/2।



MODEL EXAM-KHATA SOLUTION

Given: $L = \lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right)^{\tan x}$

Form Check: As $x \rightarrow 0^+$, $\frac{1}{x} \rightarrow \infty$ and $\tan x \rightarrow 0$. This is the indeterminate power form $[\infty^0]$.

Step 1: Take Natural Logarithm on Both Sides

Let $y = \left(\frac{1}{x}\right)^{\tan x}$.

$$\ln y = \ln \left[\left(\frac{1}{x}\right)^{\tan x} \right] = \tan x \cdot \ln \left(\frac{1}{x}\right) = \tan x \cdot (-\ln x) = -\tan x \ln x$$

Now compute the limit of the exponent:

$$k = \lim_{x \rightarrow 0^+} \ln y = - \lim_{x \rightarrow 0^+} (\tan x \ln x) \quad [0 \cdot (-\infty) \text{ form}]$$

Step 2: Transform to Quotient and Apply L'Hôpital's Rule

Rewrite product as a fraction:

$$k = - \lim_{x \rightarrow 0^+} \frac{\ln x}{\cot x} \quad \left[\frac{-\infty}{\infty} \text{ form} \right]$$

Applying L'Hôpital's Rule (differentiating numerator and denominator):

$$k = - \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(\cot x)} = - \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\csc^2 x} = \lim_{x \rightarrow 0^+} \frac{\sin^2 x}{x}$$

Since $\sin x \sim x$ as $x \rightarrow 0$:

$$k = \lim_{x \rightarrow 0^+} \left(\frac{\sin x}{x} \cdot \sin x \right) = 1 \cdot 0 = 0$$

Step 3: Exponentiate to Obtain Final Limit

$$\lim_{x \rightarrow 0^+} \ln y = 0 \implies L = \lim_{x \rightarrow 0^+} y = e^0 = 1$$

● সহজ বাংলায় বুঝি

যেহেতু পাওয়ার আকারে ∞^0 আছে, তাই প্রথমে উভয়পাশে $\ln$ ধরে নিতে হয়। এরপর গুণফলকে $\frac{\ln x}{\cot x}$ আকারে সাজিয়ে ল' হসপিটাল নিয়ম প্রয়োগ করলে ঘাতের সীমা আসে 0। পরিশেষে $e^0 = 1$ হলো চূড়ান্ত উত্তর।



MODEL EXAM-KHATA SOLUTION

Given: $L = \lim_{x \rightarrow \infty} (e^x + 1)^{-2/x}$

Form Check: As $x \rightarrow \infty$, $e^x + 1 \rightarrow \infty$ and $-2/x \rightarrow 0$. This is $[\infty^0]$ form.

Step 1: Take Logarithm

Let $y = (e^x + 1)^{-2/x}$.

$$\ln y = -\frac{2}{x} \ln(e^x + 1) = \frac{-2 \ln(e^x + 1)}{x}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{-2 \ln(e^x + 1)}{x} \quad \left[\frac{-\infty}{\infty} \text{ form} \right]$$

Step 2: Apply L'Hôpital's Rule

Differentiating numerator and denominator w.r.t x :

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{-2 \cdot \frac{e^x}{e^x + 1}}{1} = -2 \lim_{x \rightarrow \infty} \frac{e^x}{e^x + 1} = -2 \lim_{x \rightarrow \infty} \frac{1}{1 + e^{-x}} = -2(1) = -2$$

Step 3: Exponentiate to Get L

$$\lim_{x \rightarrow \infty} \ln y = -2 \implies L = e^{-2} = \frac{1}{e^2}$$

● সহজ বাংলায় বুঝি

উভয়পাশে $\ln$ নিলে হর হয় x এবং লব হয় $-2 \ln(e^x + 1)$ । ল' হসপিটাল করলে $\frac{-2e^x}{e^x + 1}$ থাকে, যার অসীম সীমা -2 । অতএব চূড়ান্ত মান e^{-2} ।

Problems 1 to 9 (Essential Classroom & Solved Notes Set)

No.	Question	Indeterminate Form	Key Calculation Method	Final Answer
1	$\lim_{x \rightarrow 0} \frac{x e^x - \ln(1+x)}{x^2}$	0/0	Apply L'Hôpital twice: $\frac{(1+x)e^x - \frac{1}{1+x}}{2x} \rightarrow \frac{(2+x)e^x + \frac{1}{(1+x)^2}}{2} = \frac{2+1}{2}$	3/2
2	$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x - 4x}{x^6}$	0/0	Series Expansion: $e^x - e^{-x} = 2x + \frac{x^3}{3} + \frac{x^5}{60}$, $2 \sin x = 2x - \frac{x^3}{3} + \frac{x^5}{60}$. Numerator = $\frac{x^5}{30}$.	1/30
3	$\lim_{\theta \rightarrow \pi/4} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$	0/0	Trig Identity: $\cos \theta + \sin \theta = \sqrt{2} \cos(\theta - \pi/4)$. Let $h = \theta - \pi/4$: $\frac{\sqrt{2}(1 - \cos h)}{16h^2} = \frac{\sqrt{2}}{16} \cdot \frac{1}{2}$	$\frac{\sqrt{2}}{32}$
4	$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a}$	0/0	Difference of powers factorization or L'Hôpital: $\lim_{x \rightarrow a} \frac{nx^{n-1}}{1} = na^{n-1}$	na^{n-1}
5	$\lim_{x \rightarrow 3} \frac{x^3 - 9}{x - 3}$	0/0	Factorization: $\lim_{x \rightarrow 3} (x + 3) = 6$	6
6	$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$	0/0	Rationalization: $\frac{(1+x)-1}{x(\sqrt{1+x}+1)} = \frac{1}{\sqrt{1+x}+1} \rightarrow \frac{1}{2}$	1/2
7	$\lim_{x \rightarrow a} \frac{x^{7/2} - a^{7/2}}{\sqrt{x} - \sqrt{a}}$	0/0	Put $t = \sqrt{x}$: $\lim_{t \rightarrow \sqrt{a}} \frac{t^7 - (\sqrt{a})^7}{t - \sqrt{a}} = 7(\sqrt{a})^6 = 7a^3$	$7a^3$
8	$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$	0/0	Identity: $\frac{2 \sin^2(x/2)}{x^2} = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2 \rightarrow \frac{1}{2}$	1/2

No.	Question	Indeterminate Form	Key Calculation Method	Final Answer
9	$\lim_{x \rightarrow 0} x^2 \log(x^2)$	$0 \cdot (-\infty)$	Rewrite as $\frac{\ln(x^2)}{1/x^2}$, apply L'Hôpital: $\frac{2/x}{-2/x^3} = -x^2 \rightarrow 0$	0

Topic 5: Differential Co-efficient by First Principle Method (3 Marks Guaranteed)

PYQ (SP-24)

3 MARKS

Find the differential co-efficient of $\sin x$ by the first principle method.

EASY



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = \sin x$

Required: Derive $f'(x) = \frac{d}{dx}(\sin x)$ by definition (First Principle).

Formula: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Step 1: Setup the Limit

$$f(x+h) = \sin(x+h)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

Step 2: Apply Trigonometric Formula $\sin C - \sin D$

Using $\sin C - \sin D = 2 \cos\left(\frac{C+D}{2}\right) \sin\left(\frac{C-D}{2}\right)$:

$$\sin(x+h) - \sin x = 2 \cos\left(\frac{x+h+x}{2}\right) \sin\left(\frac{x+h-x}{2}\right) = 2 \cos\left(x + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right)$$

Step 3: Evaluate the Standard Limit

$$f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(x + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right)}{h} = \lim_{h \rightarrow 0} \cos\left(x + \frac{h}{2}\right) \cdot \lim_{h \rightarrow 0} \frac{\sin(h/2)}{h/2}$$

Since $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$ (where $u = h/2 \rightarrow 0$ as $h \rightarrow 0$):

$$f'(x) = \cos(x+0) \cdot (1) = \cos x$$

সহজ বাংলায় বুঝি

$\sin(x+h) - \sin x$ -কে $\sin C - \sin D$ -এর ত্রিকোণমিতিক সূত্রে ভেঙে পাই $2 \cos\left(x + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right)$ । নিচে h -কে 2 দিয়ে ভাগ করে সাজালে $\frac{\sin(h/2)}{h/2} \rightarrow 1$ হয়, আর বাকি থাকে $\cos x$ ।



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = e^x$ Formula: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Step 1: Setup and Factorization

$$f(x+h) = e^{x+h} = e^x \cdot e^h$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h}$$

Step 2: Factor out e^x and Use Standard LimitSince e^x is independent of h :

$$f'(x) = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

Using the standard limit $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$:

$$f'(x) = e^x \cdot 1 = e^x$$

● সহজ বাংলায় বুঝি

লব থেকে e^x কমন নিলে ব্র্যাকেটে থাকে $\frac{e^h-1}{h}$ । আদর্শ লিমিটের সূত্রানুসারে $\lim_{h \rightarrow 0} \frac{e^h-1}{h} = 1$ । তাই গুণফল সরাসরি e^x থাকে।



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = \ln x$ ($x > 0$)

Step 1: Setup and Log Difference

$$f(x+h) = \ln(x+h)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h}$$

Using logarithmic identity $\ln A - \ln B = \ln(A/B)$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\ln\left(\frac{x+h}{x}\right)}{h} = \lim_{h \rightarrow 0} \frac{\ln\left(1 + \frac{h}{x}\right)}{h}$$

Step 2: Multiply and Divide Denominator by x

$$f'(x) = \lim_{h \rightarrow 0} \frac{\ln\left(1 + \frac{h}{x}\right)}{\frac{h}{x} \cdot x} = \frac{1}{x} \lim_{h \rightarrow 0} \frac{\ln\left(1 + \frac{h}{x}\right)}{\frac{h}{x}}$$

Let $u = \frac{h}{x}$. As $h \rightarrow 0$, $u \rightarrow 0$.

Using standard limit $\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1$:

$$f'(x) = \frac{1}{x} \cdot 1 = \frac{1}{x}$$

● সহজ বাংলায় বুঝি

$\ln(x+h) - \ln x = \ln(1 + h/x)$! নিচে x দিয়ে ভাগ ও গুণ করে আদর্শ লিমিট $\frac{\ln(1+u)}{u} \rightarrow 1$ বানালে সামনে শুধু $\frac{1}{x}$ বাকি থাকে।

Topic 6: Advanced Differentiation & Related Rates

The radius of a circle increases at a rate of 3 cm/sec . Find the rate of change of the area when: (i) radius is 5 cm , (ii) area is $4\pi \text{ cm}^2$.

EASY



MODEL EXAM-KHATA SOLUTION

Given:

- Rate of change of radius: $\frac{dr}{dt} = 3 \text{ cm/sec}$
- Area of circle: $A = \pi r^2$

Required: Find $\frac{dA}{dt}$ when (i) $r = 5 \text{ cm}$, (ii) $A = 4\pi \text{ cm}^2$.

Step 1: Differentiate Area with Respect to Time t

$$A = \pi r^2$$

Differentiating both sides with respect to time t using the Chain Rule:

$$\frac{dA}{dt} = \frac{d}{dr}(\pi r^2) \cdot \frac{dr}{dt} = 2\pi r \frac{dr}{dt}$$

Step 2: Case (i) When Radius $r = 5 \text{ cm}$

Substitute $r = 5$ and $\frac{dr}{dt} = 3$:

$$\frac{dA}{dt} = 2\pi(5)(3) = 30\pi \text{ cm}^2/\text{sec} \approx 94.25 \text{ cm}^2/\text{sec}$$

Step 3: Case (ii) When Area $A = 4\pi \text{ cm}^2$

First find the corresponding radius r :

$$\pi r^2 = 4\pi \implies r^2 = 4 \implies r = 2 \text{ cm} \quad (\text{since } r > 0)$$

Now substitute $r = 2$ and $\frac{dr}{dt} = 3$:

$$\frac{dA}{dt} = 2\pi(2)(3) = 12\pi \text{ cm}^2/\text{sec} \approx 37.70 \text{ cm}^2/\text{sec}$$

সহজ বাংলায় বুঝি

বৃত্তের ক্ষেত্রফল $A = \pi r^2$ । সময় t -এর সাপেক্ষে অন্তরীকরণ করলে চেইন রুলে $2\pi r \frac{dr}{dt}$ পাই। (i) $r = 5$ এবং $\frac{dr}{dt} = 3$ বসালে ক্ষেত্রফল বৃদ্ধির হার 30π ।
(ii) ক্ষেত্রফল 4π হলে ব্যাসার্ধ $r = \sqrt{4} = 2 \text{ cm}$, তখন হার 12π । একক cm^2/sec লেখা আবশ্যিক!



MODEL EXAM-KHATA SOLUTION

Given: Let $u = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ and $v = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$.

Required: Find $\frac{du}{dv} = \frac{du/dx}{dv/dx}$.

Step 1: Trigonometric Substitution

Substitute $x = \tan \theta \Rightarrow \theta = \tan^{-1} x$.

For u :

$$u = \tan^{-1}\left(\frac{2 \tan \theta}{1 - \tan^2 \theta}\right) = \tan^{-1}(\tan 2\theta) = 2\theta = 2 \tan^{-1} x$$

For v :

$$v = \sin^{-1}\left(\frac{2 \tan \theta}{1 + \tan^2 \theta}\right) = \sin^{-1}(\sin 2\theta) = 2\theta = 2 \tan^{-1} x$$

Step 2: Differentiate

Notice that $u = v$! Therefore:

$$\frac{du}{dv} = \frac{d}{dv}(v) = 1$$

(Or by derivatives w.r.t x : $\frac{du}{dx} = \frac{2}{1+x^2}$ and $\frac{dv}{dx} = \frac{2}{1+x^2}$, so $\frac{du}{dv} = \frac{2/(1+x^2)}{2/(1+x^2)} = 1$).

সহজ বাংলায় বুঝি

$x = \tan \theta$ বসালে ত্রিকোণমিতির আদর্শ সূত্রে উভয় রাশিই $2\theta = 2 \tan^{-1} x$ হয়ে যায়! অর্থাৎ u আর v আসলে হুবহু একই ফাংশন। একটির সাপেক্ষে অপরটির অন্তরজ সরাসরি 1।



If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$, show that $y_1 = \frac{\cos x}{2y-1}$.

MODEL EXAM-KHATA SOLUTION

Given: $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$

Step 1: Replace Infinite Tail with y

Since the square root chain extends to infinity, replacing the self-similar inner part with y gives:

$$y = \sqrt{\sin x + y}$$

Step 2: Square and Differentiate Implicitly

Squaring both sides:

$$y^2 = \sin x + y$$

Differentiating both sides with respect to x :

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(\sin x) + \frac{d}{dx}(y)$$

$$2y \frac{dy}{dx} = \cos x + \frac{dy}{dx}$$

Step 3: Solve for $y_1 = \frac{dy}{dx}$

$$2yy_1 - y_1 = \cos x \implies y_1(2y - 1) = \cos x$$

$$y_1 = \frac{\cos x}{2y - 1}$$

● সহজ বাংলায় বুঝি

অসীম রুটের ভেতরের অংশকে আবার y লেখা যায়, ফলে সমীকরণটি দাঁড়ায় $y^2 = \sin x + y$ । উভয়পাশে x -এর সাপেক্ষে ডিফারেনশিয়েট করে y_1 কমন নিলেই কাঙ্ক্ষিত প্রমাণ চলে আসে।



Exam-Khata Model Answers: Part 3

Successive Differentiation & Leibniz Theorem, Rolle's & Mean Value Theorems, and Series Expansions.

Topic 7: Successive Differentiation & Leibniz's Theorem (Guaranteed 6 Marks in Q3a)



MODEL EXAM-KHATA SOLUTION

Given: $y = x^n \log x = x^n \ln x$

Required: Prove that $y_{n+1} = \frac{n!}{x}$.

Formula (Leibniz's Theorem): $(uv)_m = u_m v + \binom{m}{1} u_{m-1} v_1 + \binom{m}{2} u_{m-2} v_2 + \dots$

Step 1: First Differentiation (y_1)

Differentiating $y = x^n \ln x$ with respect to x using the product rule:

$$y_1 = \frac{d}{dx}(x^n \ln x) = nx^{n-1} \ln x + x^n \cdot \frac{1}{x}$$

$$y_1 = nx^{n-1} \ln x + x^{n-1}$$

Multiply both sides by x :

$$xy_1 = nx^n \ln x + x^n$$

Since $y = x^n \ln x$, substitute y on the right side:

$$xy_1 = ny + x^n$$

Step 2: Apply Leibniz's Theorem n Times

Differentiating both sides n times with respect to x using Leibniz's Theorem:

$$\frac{d^n}{dx^n}(xy_1) = n \frac{d^n}{dx^n}(y) + \frac{d^n}{dx^n}(x^n)$$

For the left side $(xy_1)_n$, let $u = y_1$ and $v = x$:

$$(xy_1)_n = (y_1)_n \cdot x + \binom{n}{1} (y_1)_{n-1} \cdot \frac{d}{dx}(x) + \binom{n}{2} (y_1)_{n-2} \cdot 0$$

$$(xy_1)_n = y_{n+1} \cdot x + ny_n \cdot 1 = xy_{n+1} + ny_n$$

Step 3: Differentiate the Right Side

$$n \frac{d^n}{dx^n}(y) = ny_n$$

$$\frac{d^n}{dx^n}(x^n) = n!$$

Equating left and right sides:

$$xy_{n+1} + ny_n = ny_n + n!$$

Step 4: Cancel ny_n and Solve for y_{n+1}

$$xy_{n+1} = n! \implies y_{n+1} = \frac{n!}{x}$$

সহজ বাংলায় বুঝি

প্রথমে একবার ডিফারেনশিয়েট করে সমীকরণটিকে x -দ্বারা গুণ করে $xy_1 = ny + x^n$ সাজাতে হয়। এরপর উভয়পাশে লাইবনিজ উপপাদ্য দিয়ে n -তম অন্তরীকরণ করলে উভয় পাশ থেকে ny_n কেটে যায় এবং বামে xy_{n+1} ও ডানে $n!$ থাকে। ফলে সহজেই $y_{n+1} = \frac{n!}{x}$ প্রমাণিত হয়।

⚠ Common Exam Mistake

অনেক শিক্ষার্থী সরাসরি $y = x^n \ln x$ -এ $n + 1$ বার লাইব্রেরি প্রয়োগ করতে গিয়ে বিস্তৃতি অনেক লম্বা করে ফেলে এবং গণনা ভুল করে। $xy_1 = ny + x^n$ ট্রিকটি ব্যবহার করলে মাত্র ৪ লাইনে সমাধান হয়ে যায়!



MODEL EXAM-KHATA SOLUTION

Given: $y = \tan^{-1} x$ Required: Prove $(1 + x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$.

Step 1: First and Second Derivatives

Differentiating y with respect to x :

$$y_1 = \frac{1}{1+x^2} \Rightarrow (1+x^2)y_1 = 1$$

Differentiating again with respect to x :

$$(1+x^2)y_2 + 2xy_1 = 0$$

Step 2: Apply Leibniz's Theorem $(n-1)$ TimesWe need an equation involving y_{n+1} . Since our base equation contains y_2 , differentiating it $(n-1)$ times will produce

$$y_{2+(n-1)} = y_{n+1}.$$

Applying Leibniz's theorem $(n-1)$ times to $(1+x^2)y_2 + 2xy_1 = 0$:Term 1: $[(1+x^2)y_2]_{n-1}$ Let $u = y_2$ and $v = 1+x^2$:

$$\begin{aligned} &= (y_2)_{n-1}(1+x^2) + \binom{n-1}{1}(y_2)_{n-2} \frac{d}{dx}(1+x^2) + \binom{n-1}{2}(y_2)_{n-3} \frac{d^2}{dx^2}(1+x^2) \\ &= y_{n+1}(1+x^2) + (n-1)y_n(2x) + \frac{(n-1)(n-2)}{2}y_{n-1}(2) \\ &= (1+x^2)y_{n+1} + 2(n-1)xy_n + (n-1)(n-2)y_{n-1} \end{aligned}$$

Term 2: $[2xy_1]_{n-1}$ Let $u = y_1$ and $v = 2x$:

$$\begin{aligned} &= (y_1)_{n-1}(2x) + \binom{n-1}{1}(y_1)_{n-2} \frac{d}{dx}(2x) \\ &= y_n(2x) + (n-1)y_{n-1}(2) = 2xy_n + 2(n-1)y_{n-1} \end{aligned}$$

Step 3: Combine Terms and Simplify

$$(1+x^2)y_{n+1} + [2(n-1)x + 2x]y_n + [(n-1)(n-2) + 2(n-1)]y_{n-1} = 0$$

- Coefficient of y_n : $2nx - 2x + 2x = 2nx$
- Coefficient of y_{n-1} : $(n-1)[(n-2) + 2] = (n-1)(n) = n(n-1)$

● সহজ বাংলায় বুঝি

$y_1 = \frac{1}{1+x^2}$ থেকে বজ্রগুণন করে $(1+x^2)y_1 = 1$ পাই। আরেকবার ডিফারেনশিয়েট করলে পাই $(1+x^2)y_2 + 2xy_1 = 0$ । যেহেতু সর্বোচ্চ পদ y_{n+1} প্রয়োজন, তাই এটিকে $(n-1)$ বার লাইবনিজ দিয়ে অন্তরীকরণ করে সহগ যোগ করলেই কাঙ্ক্ষিত সমীকরণ প্রমাণিত হয়।

 PYQ (AU-25)

6 MARKS

If $x = \sin\left(\frac{1}{m} \ln y\right)$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.

HIGH



MODEL EXAM-KHATA SOLUTION

Given: $x = \sin\left(\frac{1}{m} \ln y\right) \Rightarrow \sin^{-1} x = \frac{1}{m} \ln y \Rightarrow \ln y = m \sin^{-1} x \Rightarrow y = e^{m \sin^{-1} x}$.

Required: Prove $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.

Step 1: First Derivative y_1 and Squaring

$$y = e^{m \sin^{-1} x}$$

Differentiating with respect to x :

$$y_1 = e^{m \sin^{-1} x} \cdot \frac{m}{\sqrt{1-x^2}} = \frac{my}{\sqrt{1-x^2}}$$

Multiply both sides by $\sqrt{1-x^2}$:

$$\sqrt{1-x^2}y_1 = my$$

Squaring both sides to eliminate the radical:

$$(1-x^2)y_1^2 = m^2y^2$$

Step 2: Second Derivative y_2

Differentiating both sides with respect to x :

$$\begin{aligned} \frac{d}{dx} [(1-x^2)y_1^2] &= \frac{d}{dx} [m^2y^2] \\ (1-x^2)(2y_1y_2) + (-2x)y_1^2 &= m^2(2yy_1) \end{aligned}$$

Dividing throughout by $2y_1$ (since $y_1 \neq 0$):

$$(1-x^2)y_2 - xy_1 - m^2y = 0$$

Step 3: Apply Leibniz's Theorem n Times

Differentiating $(1-x^2)y_2 - xy_1 - m^2y = 0$ n times with respect to x :

• For $[(1-x^2)y_2]_n$:

$$\begin{aligned} &= y_{n+2}(1-x^2) + \binom{n}{1}y_{n+1}(-2x) + \binom{n}{2}y_n(-2) \\ &= (1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n \end{aligned}$$

• For $[xy_1]_n$:

$$= y_{n+1}(x) + \binom{n}{1}y_n(1) = xy_{n+1} + ny_n$$

• For $[m^2y]_n$:

$$= m^2y_n$$

Step 4: Combine and Collect Like Terms

$$\begin{aligned} (1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n - [xy_{n+1} + ny_n] - m^2y_n &= 0 \\ (1-x^2)y_{n+2} - (2nx+x)y_{n+1} - [n(n-1)+n+m^2]y_n &= 0 \\ (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - [n^2-n+n+m^2]y_n &= 0 \\ (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n &= 0 \end{aligned}$$

● সহজ বাংলায় বুঝি

প্রশ্ন থেকে $y = e^{m \sin^{-1} x}$ বানিয়ে একবার ডিফারেনশিয়েট করে বর্গ করতে হয়: $(1 - x^2)y_1^2 = m^2 y^2$ । আরেকবার ডিফারেনশিয়েট করে উভয়পক্ষকে $2y_1$ দিয়ে ভাগ করলে দ্বিতীয় ক্রমের সমীকরণ পাওয়া যায়। এরপর লাইবনিজ উপপাদ্য দিয়ে n -তম বার অন্তরীকরণ করলে নিখুঁত প্রমাণ মেলে।

Topic 8: Rolle's & Lagrange's Mean Value Theorems (Guaranteed 4 Marks in Q3b)

● PYQ (SP-24)

4 MARKS

Determine whether Rolle's Theorem can be applied to $f(x) = x^4 - 2x^2$ on the closed interval $[-2, 2]$. And if so, find all values of c that satisfy the theorem.

MEDIUM



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = x^4 - 2x^2$ on $[-2, 2]$.

Step 1: Verify the Three Hypotheses of Rolle's Theorem

- Continuity:** $f(x)$ is a polynomial function of degree 4, hence it is continuous on the closed interval $[-2, 2]$.
- Differentiability:** Its derivative $f'(x) = 4x^3 - 4x$ exists for all $x \in (-2, 2)$, hence $f(x)$ is differentiable on the open interval $(-2, 2)$.
- Equal Endpoints:** Evaluate $f(x)$ at $a = -2$ and $b = 2$:

$$f(-2) = (-2)^4 - 2(-2)^2 = 16 - 2(4) = 16 - 8 = 8$$

$$f(2) = (2)^4 - 2(2)^2 = 16 - 8 = 8$$

$$\therefore f(-2) = f(2) = 8$$

Since all three conditions are satisfied, Rolle's Theorem is **applicable**.

Step 2: Find all Values of $c \in (-2, 2)$ Such That $f'(c) = 0$

$$f'(c) = 4c^3 - 4c = 0$$

$$4c(c^2 - 1) = 0 \implies 4c(c - 1)(c + 1) = 0$$

$$\implies c_1 = 0, \quad c_2 = 1, \quad c_3 = -1$$

Step 3: Verify Belonging to the Open Interval $(-2, 2)$

Notice that:

$$0 \in (-2, 2), \quad 1 \in (-2, 2), \quad -1 \in (-2, 2)$$

All three values lie strictly inside the open interval $(-2, 2)$.

● সহজ বাংলায় বুঝি

বহুপদী ফাংশন হওয়ায় এটি অবিচ্ছিন্ন ও অন্তরীকরণযোগ্য। দুই প্রান্তবিন্দু -2 ও 2 -তে মান সমান (৪)। অতএব রোলের উপপাদ্য প্রযোজ্য।

$f'(c) = 4c^3 - 4c = 0$ সমাধান করলে $c = 0, 1, -1$ পাওয়া যায়—যার প্রতিটিই $(-2, 2)$ ব্যবধির ভেতরে বিদ্যমান!



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = 2x^3 - 8x + 1$ on $[1, 3]$.

Step 1: Check Hypotheses of MVT

- $f(x)$ is a polynomial function, so it is continuous on $[1, 3]$.
- $f'(x) = 6x^2 - 8$ exists for all $x \in (1, 3)$, so it is differentiable on $(1, 3)$.

Thus, Lagrange's Mean Value Theorem is applicable.

Step 2: Compute Values at Endpoints

$$f(1) = 2(1)^3 - 8(1) + 1 = 2 - 8 + 1 = -5$$

$$f(3) = 2(3)^3 - 8(3) + 1 = 2(27) - 24 + 1 = 54 - 24 + 1 = 31$$

Average rate of change:

$$\frac{f(b) - f(a)}{b - a} = \frac{f(3) - f(1)}{3 - 1} = \frac{31 - (-5)}{2} = \frac{36}{2} = 18$$

Step 3: Solve $f'(c) = 18$ for $c \in (1, 3)$

$$f'(c) = 6c^2 - 8 = 18$$

$$6c^2 = 26 \implies c^2 = \frac{26}{6} = \frac{13}{3} \approx 4.333$$

$$c = \pm \sqrt{\frac{13}{3}} \approx \pm 2.0817$$

Step 4: Verify Interval Membership

Since the interval is $(1, 3)$:

$$c = +\sqrt{\frac{13}{3}} \approx 2.0817 \in (1, 3)$$

$c = -\sqrt{\frac{13}{3}} \approx -2.0817 \notin (1, 3)$ (rejected).

● সহজ বাংলায় বুঝি

প্রান্তবিন্দু ১ ও ৩ বসিয়ে গড় ঢাল বের করলাম: $\frac{31 - (-5)}{3 - 1} = 18$ । এরপর স্পর্শকের ঢাল $6c^2 - 8 = 18$ ধরলে $c = \sqrt{13/3} \approx 2.08$ পাওয়া যায়, যা $(1, 3)$ ব্যবধির মাঝে অবস্থিত। ফলে উপপাদ্যটি সত্য প্রমাণিত হলো।

Topic 9: Maclaurin & Taylor Series Expansions (2–3 Marks)

PYQ (SP-24, AU-25)

2 MARKS

Find the Maclaurin Series for the function $f(x) = \log(1+x)$.

EASY



MODEL EXAM-KHATA SOLUTION

Given: $f(x) = \log(1+x) = \ln(1+x)$ Formula: $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{(4)}(0) + \dots$ Step 1: Compute Successive Derivatives at $x = 0$

$$\bullet f(x) = \ln(1+x) \implies f(0) = \ln(1) = 0$$

$$\bullet f'(x) = \frac{1}{1+x} = (1+x)^{-1} \implies f'(0) = 1$$

$$\bullet f''(x) = -(1+x)^{-2} \implies f''(0) = -1$$

$$\bullet f'''(x) = 2(1+x)^{-3} \implies f'''(0) = 2$$

$$\bullet f^{(4)}(x) = -6(1+x)^{-4} \implies f^{(4)}(0) = -6$$

Step 2: Substitute into Maclaurin's Formula

$$f(x) = 0 + x(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2) + \frac{x^4}{4!}(-6) + \dots$$

$$f(x) = x - \frac{x^2}{2} + \frac{2x^3}{6} - \frac{6x^4}{24} + \dots$$

$$f(x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

● সহজ বাংলায় বুঝি

বারবার ডিফারেনশিয়েট করে $x = 0$ বসালে সহগগুলো আসে $1, -1, 2, -6$ । ফ্যাক্টোরিয়াল দিয়ে ভাগ করলে হরগুলো হয়ে যায় $2, 3, 4$ । মনে রাখবেন, এখানে হরে কোনো ফ্যাক্টোরিয়াল থাকে না!

● PYQ (AU-25)

3 MARKS

Expand $\frac{1}{x}$ in ascending powers of $(x-2)$ by Taylor's series.

MEDIUM



Realistic University Mock Examinations

Three fully calibrated 30-mark midterm mock papers with complete model answer keys and scoring rubrics.

Mock Exam 1: Standard Spring Examination

Pattern: Spring Semester • Time: 1 Hour 30 Minutes • Total Marks: 30

● HIGHLY RECOMMENDED FOR TIMED PRACTICE

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG • DEPARTMENT OF CSE

COURSE CODE: MATH-1107 • COURSE TITLE: MATHEMATICS-I (MID TERM)

1. (Total: 10 Marks)

a) Find the Maclaurin Series for the function $f(x) = \log(1+x)$.

[2 Marks]

b) Find the domain and range of the function $f(x) = |x+3| + |x-3|$.

[2 Marks]

c) Test the differentiability of the following function at $x = 3/2$:

$$f(x) = \begin{cases} 3 + 2x, & -3/2 \leq x < 0 \\ 3 - 2x, & 0 \leq x < 3/2 \\ -3 - 2x, & x \geq 3/2 \end{cases}$$

— OR —

Test the differentiability of the function at $x = \pi/2$.

$$f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \pi/2 \\ 2 + (x - \pi/2)^2, & x \geq \pi/2 \end{cases}$$

[3 Marks]

d) Evaluate the limit using L'Hôpital's rule: $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

— OR —

Differentiate $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ with respect to $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$.

[3 Marks]

2. (Total: 10 Marks)

a) Find the differential co-efficient of $\sin x$ by the first principle method.

[3 Marks]

b) If $y = \log \sec(ax + b)^3$, find $\frac{dy}{dx}$.

[3 Marks]

c) The radius of a circle increases at a rate of 3 cm/sec . Find the rate of change of the area when: (i) radius is 5 cm . (ii) area is $4\pi \text{ cm}^2$.

[4 Marks]

3. (Total: 10 Marks)

a) If $y = x^n \log x$, prove that $y_{n+1} = \frac{n!}{x}$.

[6 Marks]

b) Determine whether Rolle's Theorem can be applied to $f(x) = x^4 - 2x^2$ on $[-2, 2]$. If so, find all values of c in the interval that satisfy the theorem.

— OR —

Verify the Mean Value Theorem for $f(x) = 2x^3 - 8x + 1$ when $a = 1$ and $b = 3$.

[4 Marks]

[View Complete Step-by-Step Model Solutions & Scoring Rubric for Mock Exam 1](#)

FULL 30 MARKS KEY



Mock Exam 2: Standard Autumn Examination

AUTUMN BLUEPRINT PRACTICE

Pattern: Autumn Semester • Time: 1 Hour 30 Minutes • Total Marks: 30

1. (Total: 10 Marks)

a) Expand $\frac{1}{x}$ in ascending powers of $(x - 2)$ using Taylor's series.

[2 Marks]

b) Find the domain and range of the function $f(x) = \sqrt{x^2 - 9}$.

[2 Marks]

c) Test the continuity of the function $f(x) = \begin{cases} \frac{1 - \cos x}{x^3}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $x = 0$.

[3 Marks]

d) Evaluate the limit using L'Hôpital's rule: $\lim_{x \rightarrow \infty} (e^x + 1)^{-2/x}$.

[3 Marks]

2. (Total: 10 Marks)

a) By First Principle method, derive the differential co-efficient of $f(x) = \ln x$.

[3 Marks]

b) If $y = (\cos x)^{(\cos x)^{(\cos x)^{\dots \infty}}}$, find $\frac{dy}{dx}$.

[3 Marks]

c) Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$.

[4 Marks]

3. (Total: 10 Marks)

a) If $x = \sin\left(\frac{1}{m} \ln y\right)$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.

[6 Marks]

b) Verify the Mean Value Theorem for $f(x) = \frac{x^3}{4} + 1$ over the interval $[0, 2]$.

[4 Marks]

[View Complete Step-by-Step Model Solutions & Key for Mock Exam 2](#)

FULL 30 MARKS KEY



Mock Exam 3: High-Yield Challenger Midterm

ADVANCED SIMULATION

Designed with Highest-Probability Modified Questions • Time: 1 Hour 30 Minutes • Total Marks: 30

1. (Total: 10 Marks)

- a) Find the Maclaurin Series generated by $f(x) = \cos x$. [2 Marks]
- b) Find the domain and range of $f(x) = \sqrt{\frac{1-x}{x}}$. [2 Marks]
- c) Test the differentiability of the function at $x = 2$:
 $f(x) = \begin{cases} 3x + 2, & x > 2 \\ 12 - x^2, & x \leq 2 \end{cases}$ [3 Marks]
- d) Evaluate using L'Hôpital's rule: $\lim_{x \rightarrow 0^+} (1/x)^{\tan x}$. [3 Marks]

2. (Total: 10 Marks)

- a) By First Principle method, derive the differential co-efficient of $f(x) = e^x$. [3 Marks]
- b) If $y = \sqrt{\sin x} + \sqrt{\sin x} + \dots \infty$, show that $y_1 = \frac{\cos x}{2y-1}$. [3 Marks]
- c) The radius of a circle increases at a rate of **3 cm/sec**. Find the rate of change of the area when: (i) radius is **5 cm**, (ii) area is **$4\pi \text{ cm}^2$** . [4 Marks]

3. (Total: 10 Marks)

- a) If $y = \tan^{-1} x$, show that $(1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$. [6 Marks]
- b) Verify the Mean Value Theorem for $f(x) = 3 + 2x - x^2$ in the interval $(0, 1)$. [4 Marks]

[View Complete Step-by-Step Model Solutions & Key for Mock Exam 3](#)

FULL 30 MARKS KEY



High-Probability PYQ Question Patterns

Observed structural recurrence patterns from recent IIUC examinations.

Pattern 1: The Leibniz Differential Equation (6 Marks)

EXTREMELY REPEATED

Archetype: Given $y = f(x)$, find a relation between y_2, y_1, y , and apply Leibniz's theorem n times to form an equation with y_{n+2}, y_{n+1}, y_n .

Dominant Variants:

- **Math input error** (Sp-24)
- $y = an^{-1}x \implies (1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$ (Sp-25)
- **Math input error** (Au-25)
- $y = e^{a \sin^{-1} x} \implies (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$

Exam Probability: *Extremely High based on observed pattern.*

Pattern 2: The Expanding Circle Related Rates (4 Marks)

EXTREMELY REPEATED

Archetype: "The radius of a circle increases at a rate of **3extcm/sec**. Find the rate of change of the area when (i) radius is **5extcm**, (ii) area is **4nextcm²**."

Why Important: Featured identically in the teacher's formal Question Sheet, in the lecture notes, and in the Spring 2024 examination paper! Appears as Q2(c).

Exam Probability: *Very High probability based on observed course consistency.*

Pattern 3: Rolle's or MVT Verification on Cubic/Quartic (4 Marks)

HIGHLY REPEATED

Archetype: Given a polynomial $f(x)$ on $[a, b]$, verify hypotheses (continuity, differentiability, and $f(a) = f(b)$ for Rolle), then solve for $c \in (a, b)$.

Dominant Variants:

- Rolle: $f(x) = x^4 - 2x^2$ on $[-2, 2]$ (Sp-24), $f(x) = 2x^3 + x^2 - 4x - 2$ on $[-\sqrt{2}, \sqrt{2}]$ (Sp-25), $f(x) = x^2 - 3x + 2$ on $[1, 2]$ (Au-25).
- MVT: $f(x) = 2x^3 - 8x + 1$ on $[1, 3]$ (Sp-24), $f(x) = 3 + 2x - x^2$ on $[0, 1]$ (Sp-25), **Math input error** on $[0, 2]$ (Au-25).

Exam Probability: *Guaranteed to be asked in Question 3(b).*

Pattern 4: Trigonometric Substitution Derivatives (3 Marks)

FREQUENTLY TESTED

Archetype: Differentiating one inverse trigonometric function with respect to another.

Key Example: Differentiate $\sin^{-1}x$ with respect to $\cos^{-1}x$. Both reduce to $\frac{d}{dx} \sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$, so the derivative is simply 1!

Exam Probability: High probability alternative in Question 1(d).



Expected Practice Question Bank

Tiered practice questions curated strictly from the analyzed course repository.

● Must Solve (Non-Negotiable)

Q No.	Question Statement	Topic	Marks	Source / Tag
1	If $y = x^n \log x$, prove that $\frac{dy}{dx} = nx^{n-1} \log x + x^{n-1}$.	Leibniz Theorem	6M	Sp-24, Lecture Notes
2	If $y = ax^{-1}x$, show $(1+x^2)y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$.	Leibniz Theorem	6M	Sp-25, Lecture Notes
3	Circle radius increases at $3\pi \text{ cm/s}$. Find rate of area change when: (i) $r = 5\pi \text{ cm}$, (ii) $A = 4\pi \text{ cm}^2$.	Related Rates	4M	Sp-24, Q-Sheet
4	Determine whether Rolle's theorem applies to $f(x) = x^4 - 2x^2$ on $[-2, 2]$ and find c.	Rolle's Theorem	4M	Sp-24, Q-Sheet
5	Verify MVT for $f(x) = 2x^3 - 8x + 1$ when $a = 1, b = 3$.	Mean Value Theorem	4M	Sp-24, Q-Sheet
6	Find differential coefficient of $\sin x$ by First Principle method.	First Principle	3M	Sp-24, Handwritten Notes
7	Find differential coefficient of e^x by First Principle method.	First Principle	3M	Sp-25, Handwritten Notes
8	Evaluate using L'Hôpital: $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.	Indeterminate Form	3M	Sp-24, Notes
9	Evaluate using L'Hôpital: $\lim_{x \rightarrow 0} (1/x)^{ax}$.	Indeterminate Form	3M	Sp-25, Notes
10	Find domain and range of $f(x) = x+3 + x-3 $.	Domain & Range	2M	Sp-24, Q-Sheet
11	Find Maclaurin series for $f(x) = \log(1+x)$.	Maclaurin Series	2M	Sp-24, Au-25

● Should Solve (High-Value Buffer)

Q No.	Question Statement	Topic	Marks	Source / Tag
12	If $y = ax^{-1}x$ show $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$.	Leibniz Theorem	6M	Au-25, Q-Sheet
13	Verify Rolle's theorem for $f(x) = 2x^3 + x^2 - 4x - 2$ over $[-\sqrt{2}, \sqrt{2}]$.	Rolle's Theorem	4M	Sp-25, Q-Sheet
14	Verify MVT for $f(x) = 3 + 2x - x^2$ in $(0, 1)$.	Mean Value Theorem	4M	Sp-25, Q-Sheet
15	Verify MVT for $\sin^{-1}x$ on $[0, 2]$.	Mean Value Theorem	4M	Au-25, Q-Sheet
16	Differentiate $f(x) = \ln x$ by First Principle.	First Principle	3M	Au-25, Notes
17	Test differentiability of $f(x)$ at $x = 3/2$ ($3 + 2x/3 - 2x/-3 - 2x$).	Differentiability	3M	Sp-24, Notes

Q No.	Question Statement	Topic	Marks	Source / Tag
18	Test differentiability of $f(x)$ at $x = \pi/2$ ($1/1 + \sin x/2 + (x - \pi/2)^2$).	Differentiability	3M	Sp-24, Notes
19	Test differentiability of $f(x)$ at $x = 2$ ($3x + 2; x > 2$ and $12 - x^2; x \leq 2$).	Differentiability	3M	Sp-25, Notes
20	Differentiate $\frac{1}{\sqrt{1-x^2}}$ w.r.t x .	Diff. w.r.t Function	3M	Sp-24, Sp-25
21	If $y = \sqrt{\sin x + \sqrt{\sin x + \dots}}$, show $\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}$.	Nested Radical	3M	Sp-25, Q-Sheet
22	Find domain and range of $f(x) = \sqrt{x^2 - 9}$ and $\cos^{-1}(x)$.	Domain & Range	2M	Au-25, Sp-25

Last-Minute Revision System

Time-compressed revision roadmaps designed for rapid grade maximization.

1-Day Revision Plan (24-Hour Master Schedule)

Goal: Complete syllabus mastery, solve 15 essential questions by hand, and write Mock Exam 1 under timed conditions.

- Morning (Session 1, 3 Hours):** Master Successive Differentiation & Leibniz Theorem. Work through $y = x^n \log x$, $y = an^{-1}x$, and $\frac{1}{\sqrt{1-x^2}}$ line-by-line. [Guarantees 6M].
- Mid-Day (Session 2, 2.5 Hours):** Solve Rolle's Theorem & Lagrange MVT verification problems ($x^4 - 2x^2$, $2x^3 - 8x + 1$, $3 + 2x - x^2$). Memorize continuity and differentiability statement templates. [Guarantees 4M].
- Afternoon (Session 3, 2.5 Hours):** Master Related Rates (Circle area/radius problem) and First Principle derivatives ($\sin x$, e^x , $\ln x$). [Guarantees 7M].
- Evening (Session 4, 2.5 Hours):** Piecewise differentiability testing (boundary checks for $x = 3/2, \pi/2, 2$) and Indeterminate Forms (Problems 10, 11, 12 from sheet). [Guarantees 6M].
- Night (Session 5, 2 Hours):** Review Domain & Range rules, Maclaurin's series for $\log(1+x)$ and $\cos x$, and complete Mock Exam 1 in 90 minutes.

3-Hour Fast-Track Revision Plan (Target: 24+ Marks)

Focus strictly on the top 5 highest-yield archetypes that constitute 80% of the midterm marks:

Time Block	Target Problem / Topic	Target Marks
Hour 1 (0:00 - 1:00)	Leibniz Theorem: $\frac{1}{\sqrt{1-x^2}}$ and $y = an^{-1}x$. Write the full derivations twice!	6 MARKS
Hour 2 (1:00 - 2:00)	Rolle's Theorem ($f(x) = x^4 - 2x^2$) + Lagrange MVT ($f(x) = 2x^3 - 8x + 1$) + Circle Related Rates ($A = \pi r^2$).	8 MARKS
Hour 3 (2:00 - 3:00)	First Principle of $\sin x$ & e^x + Differentiability test at $x = 3/2$ or $\pi/2$ + Maclaurin series for $\log(1+x)$.	10 MARKS

1-Hour Emergency Revision (Quick 18+ Marks Rescue)

If exam starts in 60 minutes, memorize ONLY these 5 exact items:

- Leibniz Setup:** Remember the trick $xy_1 = ny + x^n$. Differentiate n times: $xy_{n+1} + ny_n = ny_n + n! \implies y_{n+1} = n!/x$. (6 Marks!).
- Circle Formula:** $\frac{1}{\sqrt{1-x^2}}$. For $r = 5$, ans is 30π . For $A = 4\pi$, $r = 2$, ans is 12π . (4 Marks!).
- Rolle's / MVT Equation:** For $x^4 - 2x^2$ on $[-2, 2]$, roots of $4c^3 - 4c = 0$ are $c = -1, 0, 1$. For MVT, calculate $\frac{1}{\sqrt{1-x^2}}$ and solve $f'(c) = \text{slope}$. (4 Marks!).
- First Principle $\sin x$:** $\frac{1}{\sqrt{1-x^2}}$. (3 Marks!).

5. Maclaurin $\log(1+x)$: Math input error (2 Marks!).



Final Exam Checklists

Interactive formula verification and exam-day khata strategy checklists.

Interactive Formula & Topic Mastery Checklist

- ☐ Leibniz Theorem Formula: Math input error
- ☐ Leibniz Standard Proof: I can prove Math input error flawlessly.
- ☐ Circle Related Rates: Math input error with units $\text{extcm}^2/\text{extsec}$.
- ☐ Rolle's Theorem Hypotheses: (1) Continuous on $[a, b]$, (2) Differentiable on (a, b) , (3) $f(a) = f(b)$. Solve $f'(c) = 0$.
- ☐ Mean Value Theorem: Continuous on $[a, b]$, Differentiable on (a, b) . Solve Math input error
- ☐ First Principle Definition: Math input error for $\sin x$, e^x , $\ln x$, $\log_a x$.
- ☐ Piecewise Differentiability: Check Continuity first! If continuous, compare $Lf'(a)$ and $Rf'(a)$.
- ☐ L'Hôpital Power Limit: For $\lim(1/x)^{ax}$ and $\lim(e^x + 1)^{-2/x}$, take $\ln y$ first, solve exponent, then e^{extlimit} .
- ☐ Domain & Range: Set radicand ≥ 0 , denominator $\neq 0$; express x in terms of y for Range.
- ☐ Maclaurin Series: Math input error for $\log(1+x)$ and $\cos x$.

Exam-Day Khata Writing Strategy Checklist

- ☐ Write answers sequentially as requested in exam instructions: Question 1 parts, then Question 2, then Question 3.
- ☐ Always write the standard headings: **Given:**, **Formula:**, **Solution:**, and **Therefore:**.
- ☐ Box your final answer (e.g. Math input error) so the examiner can spot it instantly.
- ☐ Never skip the limit operator $\lim_{h \rightarrow 0}$ in First Principle derivations until $h = 0$ is actually substituted.
- ☐ Verify that your value of c for Rolle or MVT strictly belongs to the open interval (a, b) .